Performance Evaluation of Proof-of-Work and Collatz Conjecture Consensus Algorithms

Hamad Mousa A. Aljassas and Sreela Sasi
Department of Computer and Information Science.
Gannon University, Erie, PA-16541, USA
Al-jassas@hotmail.com, sasi001@gannon.edu

Abstract— **Blockchain is the underlying technology of Bitcoin that allows a peer-to-peer distributed ledger with security and immutability. The core of a blockchain is the consensus mechanism that sets the rule for nodes in handling the shared data. Implementation of the consensus algorithm depends on the nature of targeted business environment. In this research, the performance of two consensus algorithms, Proof-of-Work (PoW) and Proof-of-Collatz Conjecture (PCC), are studied in the context of a private blockchain. A quantitative analysis on the execution time, deployment time, and latency time are done for 1, 10, 100, 1000, and 10000 transactions and the results are presented. The results shows that PCC takes only (1/1000)th of the execution time that is required for PoW for these different sets of transactions. In addition, these timings are recorded for ten repeated executions for the same sets of transactions, and found that PCC has a nearly consistent execution time.**

Keywords—Blockchain, Proof-of-Work, Proof-of-Collatz-Conjecture, Distributed ledger, peer-to-peer ledger

I. INTRODUCTION

Blockchain is a digital ledger, recorded chronologically, that is maintained by several nodes in peer-to-peer network. Even though, there are many different opinions about the potential of blockchain technology, the security of the technology is not a point of arguments. The concept was published in 2009 by Satoshi Nakamoto as the underlying technology of Bitcoin. He presented a purely peer-to-peer version of electronic cash that would allow online payments to be sent directly from one party to another without going through a third party. In his research, the network timestamped the transactions by hashing them into an ongoing chain of hash-based proof-of-work. This forms a record that cannot be changed without redoing the proof-of-work. Since the majority of CPU power is controlled by the nodes that are not cooperating to attack the network, they will generate the longest ledger and outpace the attackers. The network itself requires minimal structure [1]. Andreas M. Antonopoulos, a blockchain enthusiast and author of Mastering Bitcoin Book, described blockchain technology as a highly disruptive technology. He described that the potential applications of blockchain technology would allow running as a decentralized trust-less system. In addition, it will not rely on anyone as an intermediary of trust [2].

Blockchain technology is expanding the horizon of digital trust further to include already existing and new applications. Authentication is not the only factor that blockchain technology promises but also promises authorization, which is crucial to digital transactions. The blockchain ecosystem consists of four main components: nodes, application, shared ledger, consensus algorithm. The nature of the blockchain is distributed and the need for each node holding the ledger to achieve an agreement is a crucial part of the whole system. This agreement is known as consensus algorithm, which is the nucleus of the blockchain system. The block is constructed using a consensus algorithm. The nodes in a peer-to-peer network are responsible for witnessing all the transactions and checking for their validity. These transactions will be then processed into a block. The nodes in the Bitcoin system make sure that a block is valid if it achieves the network requirements. One main requirement is that the hash of the broadcasted block must contain the previous block hash as illustrated in Fig. 1. If the node includes invalid transactions, then the network will reject the block. Thus, the process being done on that block is considered wasted for that particular node. The blockchain nodes must agree on what is being added to the blockchain despite them knowing each other. The block that is added to the blockchain requires to solve a mathematical puzzle using the proof-of-work that requires CPU time and energy. Hence, the node must process only a valid transaction [3].

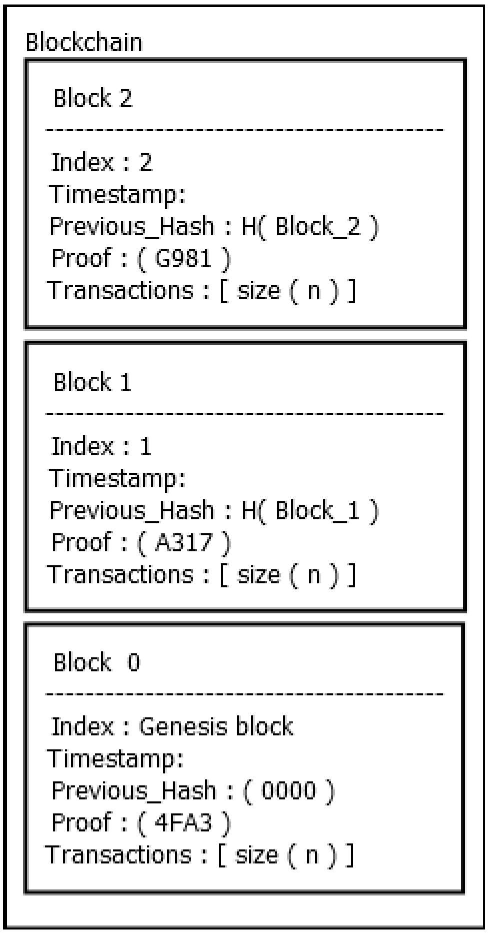

Fig. 1. Shared Blockchain

978-1-7281-0108-8/19/$31.00 ©2019 IEEE

II. BACKGROUND RESEARCH

The Proof-of-Work (PoW) consensus algorithm is a prime number-based verification mechanism that depends on a particular node to find a nonce, which is a 4 bytes field for every given block. This is done in every 10 minutes on the blockchain to prevent any possible data tempering or double-spending of transactions. The list of consensus algorithms is increasing, the major ones being Proof of Stake (PoS), Proof of Elapsed Time, and Practical Byzantine Fault Tolerance [4]. Proof of Stake utilizes the nodes' wealth for selecting the next accepted node that can add a block to the ledger. It is specially designed to solve the problem of the high power consumption of PoW [5]. This consensus has many issues since its creation and many research papers have been published in order to modify it. In [6], Vasin proposed as a system to solve the issues related to PoS security flaws. He mentioned that the PoS based systems are predictable enough to the point in which participating nodes can precompute the PoS. In [7] Luon et al. highlighted major security issues and challenges related to both PoW and PoS consensus algorithms. In [8] Zibin et al. provided a comprehensive survey of major consensus algorithms and discussed their major challenges and their recent advances. Some of the major issues discussed are selfish mining, privacy leakage and scalability. Proof of Elapsed Time, developed by Intel, is a lottery-based consensus in which the node is selected based on completing a designated waiting time. There are three main steps for adding a block to the blockchain. First, registration; where the node registers their pair of keys and waiting time. Second, calculation; where the nodes apply an equation to calculate their waiting time. Third, verification; where, whenever a block is generated by a node it will be verified by other nodes before it is accepted by the system. The basic idea is to use the z-test to check whether a node is generating blocks too fast, by winning too frequently in the competition, with other nodes for block creation [9].

The Practical Byzantine Fault Tolerance, which is used in IBM's Hyperledger Fabric consists of five steps: Request, Pre-Prepare, Prepare, Commit, and Replay. It solves the scalability issues that existed with PoW and will work only with permissioned network [10]. Collatz Orbits is documented as an alternative to Proof-of-Work, which is a number theoretic-based verification mechanism. This consensus allows each node to calculate the Collatz sequence of any given block. The iteration required to reach the stopping point, which is equal to 1, is unpredictable and will vary in time for every given integer, K [11].

The efficiency and the overall safety of a blockchain is determined by selecting a fit consensus algorithm, taking into consideration the nature of the targeted business. The blockchain is categorized into three types: Public blockchain, Private blockchain and Permissioned blockchain. The Public blockchain is accessible to everyone in the public domain. Hence, anyone can join the network as a node and contribute to it to obtain the rewards by following the network rules. However, the Private blockchain is restricted only to pre-defined nodes. The rest of the nodes may have limited access only to read. Permissioned blockchain means that the blockchain is composed of many parties and the main nodes are pre-specified by the participants. Each of these blockchains require different types of consensus [12].

III. RELATED WORK

Hao et al. conducted a recent study on the performance analysis of consensus algorithm in private blockchains, Ethereum and Hyperledger with different number of transactions. They identified the latency and throughput as the evaluation metrics. The study also illustrated the criticality of the consensus algorithm on the universal performance of private blockchain for varying number of transactions [13]. Pongnumkul et al. presented a similar research on the performance analysis of consensus algorithm in [14]. They evaluated the performance of consensus algorithms on the private blockchain platforms of Ethereum and Hyperledger Fabric. The Ethereum used PoW consensus algorithm whereas the Hyperledger Fabric used Practical Byzantine Fault Tolerance. They evaluated the execution time, latency and throughput and a comparison was provided. Their study is important in two ways: They presented a methodology for evaluating a blockchain platform. In addition, the analysis results will help the practitioners in making decisions regarding the adoption of blockchain technology in their IT systems. Both these platforms are still not competitive with the current database systems in term of performances with high workload scenarios. Proof-of-Collatz Conjecture (PCC) is introduced recently, as an alternative to PoW, which is a number based theoretic proof-of-work that includes all prime and non-prime numbers. This uses a new metric, Collatz orbits, which is associated with the Collatz Conjecture algorithm whose natural generalization is algorithmically undecidable. It exhibits properties particularly well suited to be the base for new cryptographic applications [16].

In this research, the performance analysis was conducted on the Proof-of-work consensus and the Collatz Conjecture consensus algorithms. It is done with respect to the execution time, the deployment time and the latency on a private blockchain. The consensus algorithm influences the potential of a blockchain in various areas such as stability of the ledger, speed of the network, and the fairness to the contributing nodes in the network. The investigation carried in this research leverage various operating systems and different hardware. Moreover, this research will lay the path for further scientific investigation on the increasing number of consensus algorithms. In addition, the findings will help the scientific community when implementing blockchain in various environments.

IV. ARCHITECTURE

The architecture for the Proof-of-Work and Proof-of-Collatz Conjecture algorithms consists of a network containing the transactions pool, the blockchain or ledger, and the computation nodes. The architectures are compared, and is shown in Fig. 2(a). The essential goal of a node is to add a new block into the blockchain as fast as possible. In the universe of cryptocurrency the first node to add a new block is granted the network reward, which is currently 12.5 coins in bitcoin. Initially, each node in the system obtain 'K' number of transactions from a shared pool of transactions in order to start constructing a block. The number of transactions depends on the block size defined in the blockchain platform. For instance, in the bitcoin platform, the maximum number of block size is hard coded to be 1MB. Each transaction consists of four attributes: sender, recipient, data, and signature, as illustrated in Fig. 2(b). The value of the data field in the case of cryptocurrency is the amount of coins that the sender wishes to send. Then, a node will start validating each transaction. In the bitcoin ecosystem, the transaction is valid if the sender has sufficient amount of coins based on his public key in the blockchain. The public key of the recipient is also validated. Once the transactions are valid, the node will find a special hash for these transactions using the consensus mechanisms. The hash required to meet the criteria of the network in the proof-of-work consensus mechanism, which is an adjustable 4-bytes target, is called nonce or proof. The proof is a number that will result in a hash with a valid specification required by the network. This happens when the proof is hashed with the previous block's hash and transactions. In PoW the hash must contain some leading zeros which is also known as the 'network difficulty'. In addition, the network difficulties increase as the blockchain grows. Once a hash is valid then a block is considered as constructed and is added to the existing blockchain. Then the node will distribute the new blockchain to other nodes in the system. The other nodes receiving this new blockchain will validate it by checking its contents and its hash. In this research, the Proof-of-work and Collatz-Conjecture consensus algorithms are compared using their average execution time, average deployment time and the average latency.

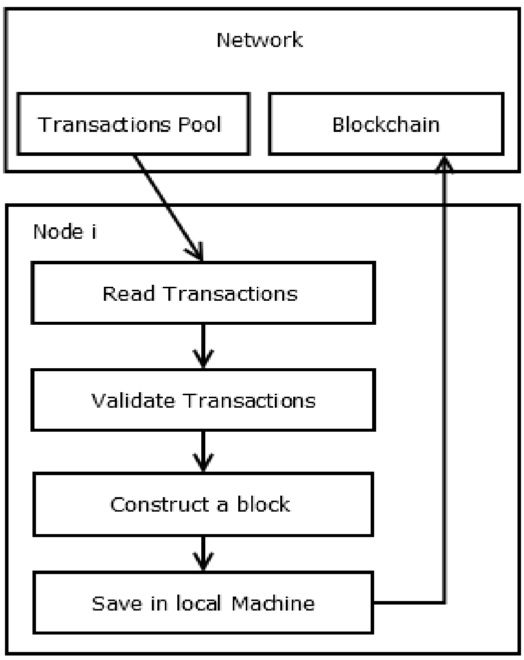

Fig. 2(a). Architecture for Comparison of PoW and PCC

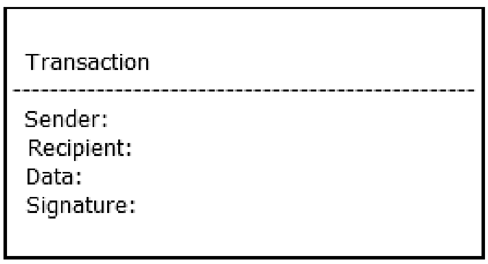

Fig. 2(b). Transaction Content

A. *Consensus Mechasnim*

The consensus mechanism plays a crucial role in the blockchain system, and is used for constructing every new block. It helps the nodes to agree on the validity of the shared blockchain or ledger despite the nodes knowing each other. The essence of blockchain's consensus is that the nodes must spend time constructing a block. The block is constructed by spending CPU time finding a unique proof using the hash function that is related to the block before it. This proof is a unique number that is obtained by finding the result of an equation or as a solution to a mathematical puzzle. The uniqueness of the proof is what makes the blockchain tamper-proof ensuring its strong security. Thus, any change in an existing block in the blockchain will require CPU time on all the blocks that comes after it as illustrated in Fig. 1.

B. *Proof-of-Work Consensus Mechasnim*

The Proof-of-Work is introduced as an economic measure to prevent denial of service attacks and other service abuses. The concept was proposed by Cynthia Dwork and Moni Naor in 1992 [15]. The main idea is that it requires a user to compute a moderately hard function in order to gain access to the resource in the network. In the Proof-of-Work blockchain model, a set of transactions, the previous block hash and a proof are hashed to produce a unique block hash that meets the network requirement. The network requires that the nodes constructing a block must have a hash that contains leading zeros. First, the proof is initialized to 0. Then the hash function is executed using the proof, and the previous block hash and the transactions to find the hash of the current block. If the hash result does not meet the network requirement, the proof is incremented and the process is repeated until the hash function yields a hash that meets the network requirements as shown in Fig. 3(a).

C. *Proof-of-Collatz Conjecture Consensus Mechansim*

The Proof-of-Collatz- Conjecture is used as the consensus mechanism for the blockchain. The time that the central processing unit spends for PCC is the time for calculating the collatz conjecture for a given block. The number of steps required to reach the target number, which is '1' in this case, in PCC is unpredictable for a varying set of transactions combined with the previous hash. Fig. 3(b) shows the PCC consensus model. The Collatz Conjecture takes the previous block hash and the current transactions existed in the transaction pool. The previous hash and the transactions will be converted into integers. The sum of these integers will be either divided or multiplied until it is equal to 1, and the process is shown in Fig. 3(b). The proof is the number of iterations that is taken to reach 1.

V. SIMULATION

The simulation is carried out on five Servers with varying resources: Memory, Processor, Operating System, and Disk Size. Three of them are with 16 GB RAM, Intel core i7-4790 CPU @ 3.60 GHz × 8. Two servers with 1 TB hard drive and 1 with 256 GB. In this, two of the servers are running on Ubuntu 17.10, and the third one is running on Windows 10. The fourth server is with 4 GB RAM, Intel core i7-2600s CPU @ 3.8 GHz × 8, with 256GB hard drive and running on Ubuntu 17.10. The fifth server is a Mac mini, with 8GB-1600 MHz DDR3, 2.5GHz Intel core i5, and 500GB SSD. The simulation is done on a private blockchain environment. In this research, synthetic transactions is used to generate 1, 10, 100, 1000, and 10,000 transactions with 1000 wallets with public key and private key using Diffie-Hellman key exchange. Initially, the Proof-of-work is used as the consensus mechanism. The network requirement is set at '2', which means that two leading zeros is necessary for a block's hash to be accepted by the network. The experiments were executed 10 times for each set of transactions 1, 10, 100, 1000 and 10,000. Then the network requirement of the hash (difficulty) is increased to '3', '4', and '5' leading zeros respectively for each of the 10 executions to gain an accurate average measurement. Similarly, the Proof-of-Collatz-Conjecture is executed with the same configuration as the Proof-of-Work. The results are given in the Table 1. The Table 1 consists of the execution time, the deployment time and the latency for difficulty = 5. The execution time is the time needed for each set of transactions to be confirmed, which is the block confirmation time that includes the deployment time. The latency is the time between the transactions deployment time and the block confirmation time for every given transaction. The average latency is the average of all repeated transactions' latency for each set.

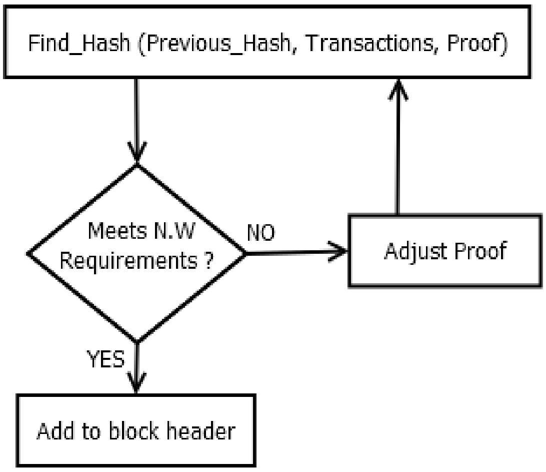

Fig. 3(a). Proof-of-Work Model

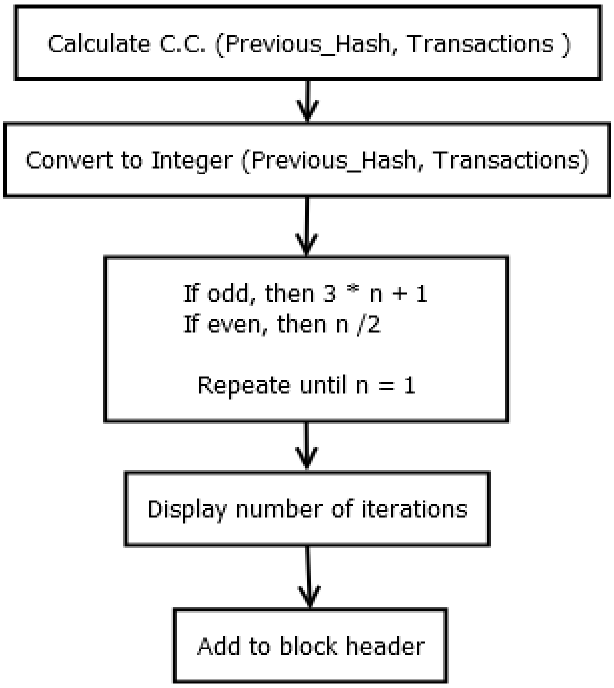

Fig. 3(b). Proof-of-Collatz-Conjecture Model

VI. RESULT

In this research, different set of transactions were experimented using Proof-of-work and Proof-of-Collatz-Conjecture. The results in terms of execution, deployment, and latency are presented in Table 1. The proof-of-work mechanism was done for a difficulty of 2, 3, 4 and 5, meaning the network requirement was set at 2, 3, 4 and 5 leading zeros for the block's hash to be valid. The average execution time per seconds for the set of 10,000 transactions with the difficulty (network requirement of the hash) = 2 is 5.94257937, with difficulty = 3 is 186.065466, with difficulty = 4 is 1689.75737, and with difficulty = 5 is 7570.703. Similar executions for a set of 1, 10, 100 and 1000 are also done. It was found that the execution time increases as the number of transactions increase as shown in Fig. 4(a). Fig. 4(b) shows the difference in execution and deployment times for PoW with difficulty 2 for different sets of transactions. Also, the Proof-of-Work took 117.7087605 seconds (difficulty = 4) for the system to find its required hash for the first time, and it took 4.127473354 seconds for another run with the same set of transactions. Hence, the Proof-of-Work algorithm has different timings for execution for the same number of transactions at different times. Fig. 4(c) shows the variation in execution timings for 100 transactions at different times using PoW. This is due to the random nature associated with the Proof-of-Work algorithm to determine the hash value required by the network. Moreover, the higher the network requirement is, the more the execution time needed as shown in Fig. 4(a). The execution times for the set of 1, 10, 100, 1000, and 10,000 transactions using Proof-of-Collatz-Conjecture are 1.173552227, 1.096043116, 1.089416538, 1.186706265 and 1.257633 respectively. A comparison of the execution and deployment times of PCC is shown in Fig. 4(d) for different sets of transactions. The Proof-of-Collatz-Conjecture shows a more stable execution time irrespective of the number of transactions. Fig. 4(e) shows the variation in execution timings using PCC for 100 transactions. The result shows that as the number of transactions increase, the execution time for PoW increases 1000 fold, whereas, for PCC, it is negligible. The deployment functionality is native for both the systems. The latency is the difference between the execution time and the deployment time.

TABEL 1. Execution time, Deployment time, and latency with Difficulty=5

Transactions	**Execution (E)**		**Deployment (D)**		**Latency (L) = E - D**	
	PoW (difficulty = 5)	***PCC***	***PoW (difficulty = 5)***	***PCC***	***PoW (difficulty = 5)***	***PCC***
1	14.21531918	1.173552227	1.033034325	1.140572309	13.18228486	0.032979918
10	52.37683799	1.096043116	0.987087011	1.06153	52.37683799	0.034513116
100	627.8809724	1.089416538	1.002723455	1.053679	627.8809724	0.035737538
1000	2743.935012	1.186706265	1.174729586	1.14018	2743.935012	0.046526265
10,000	7570.703	1.257633	1.066126108	1.170331	7570.703	0.087302

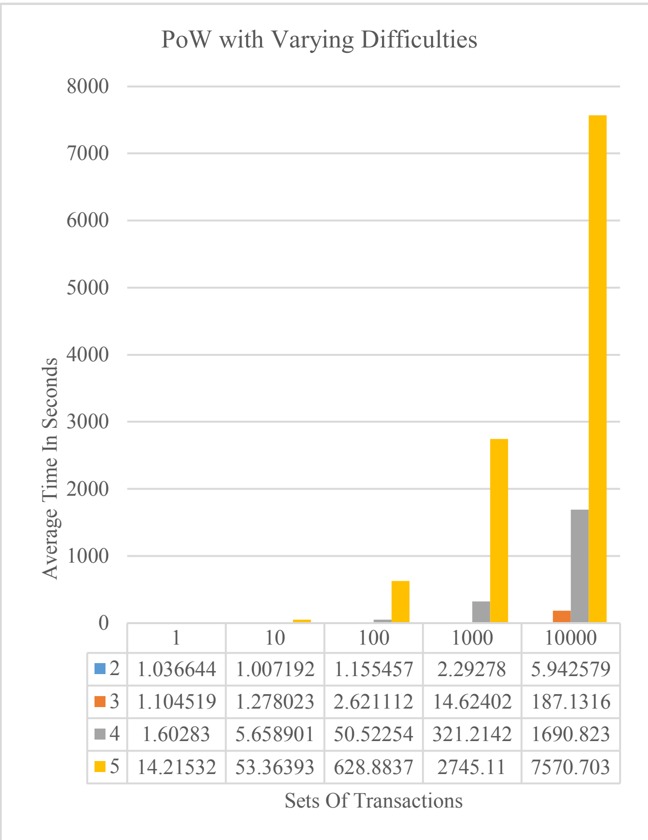

Fig. 4(a). Execution time for PoW with difficulty 2, 3, 4, and 5

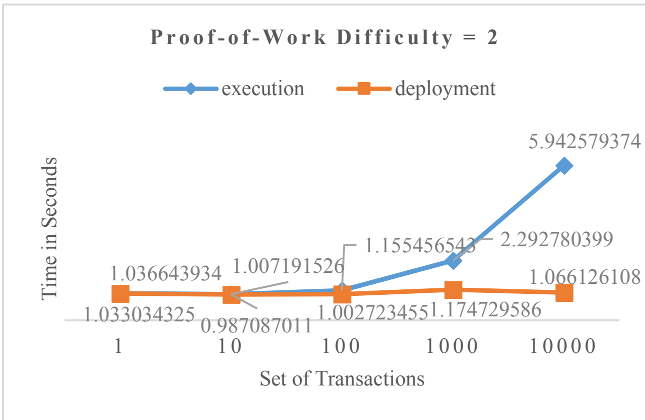

Fig. 4(b). Execution and Deployment times with Difficulty=2 for PoW

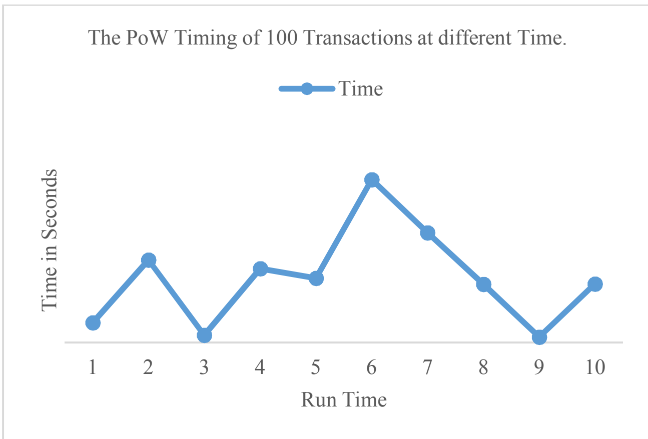

Fig. 4(c). The PoW timing for 100 transactions at different runs

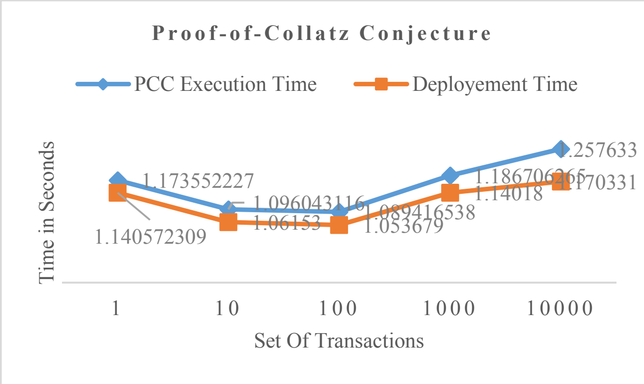

Fig. 4(d). Execution and Deployment times for PCC

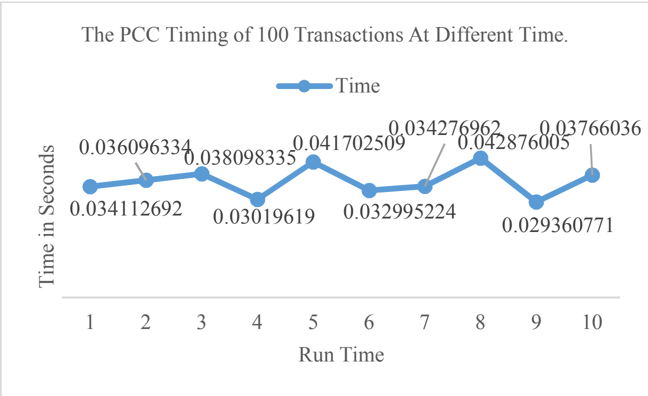

Fig. 4(e). The PCC timing for 100 transactions at different runs

VII. CONCLUSTION

The research presents a quantitative analysis on the execution, deployment and latency for Proof-of-Work (PoW) and Proof-of-Collatz-Conjecture (PCC) consensus mechanisms used in blockchain. The experiments are carried out on a private blockchain consisting of five servers. The execution and deployment times were noted for 1, 10, 100, 1000, and 10,000 transactions and they are repeated 10 times. Results show that blockchain with PCC consensus notably outperform blockchain using PoW with respect to execution time, deployment and average latency. There is a thousand fold increase in these times for PoW. In addition, it is noted that PoW gives different execution times for the same number of transactions at different times due to the random nature of finding the block's hash. However, PCC shows minimal change because it is using the same Collatz Conjecture equation. As a future work, the PCC algorithm could be tried in a public blockchain with huge number of transactions. In addition, PCC algorithm could be investigated on different applications of blockchain for its scalability and speed.

REFERENCES

[1] Nakamoto, Satoshi. "Bitcoin: A Peer-to-Peer Electronic Cash System". p.p 8. 2008.

[2] Andreas M. Antonolopous, "Mastering Bitcoin: unlocking Digital Crypto-Currencies", 2014. [E-Book] Available: Google Play Books.

[3] Taylor, David. "An Analysis of Bitcoin and the Proof of Work Protocols Energy Consumption, Growth, Impact and Sustainability," Department of Mechanical and Aerospace Engineering. 2018. Available: http://www.esru.strath.ac.uk/Documents/MSc_2018/Taylor.pdf. [Accessed Oct. 28, 2018].

[4] Baliga A. "Understanding blockchain consensus models." *Persistent*. 2017. Available: https://pdfs.semanticscholar.org/da8a/37b10bc1521a4d3de925d7ebc44bb606d740.pdf. [Accessed Sept. 12, 2018]

[5] Sunny King, Scott Nadal, "PPCoin: Peer-to-Peer Crypto-Currency with Proof-of-Stake," Aug 2012. Available: https://peercoin.net/assets/paper/peercoin-paper.pdf. [Accessed Nov. 29, 2018]

[6] Vasin Pavel, "Blackcoin's proof-of-stake protocol v2," 2014. [Online Serial]. Avaliable: *https://blackcoin.co/blackcoin-pos-protocolv2-whitepaper.pdf*. [Accessed Oct. 27, 2018].

[7] Lin, Iuon-Chang, and Tzu-Chun Liao. "A Survey of Blockchain Security Issues and Challenges." *International Journal of Network Security*, no. 5, 2017. Available: https://pdfs.semanticscholar.org/f61e/db500c023c4c4ef665bd7ed2423170773340.pdf. [Accessed Oct. 12, 2018].

[8] Zheng, Zibin, et al. "Blockchain challenges and opportunities: A survey," International Journal of Web and Grid Services, vol. 14, no. 4, 2018. Available: https://www.researchgate.net/profile/Hong-Ning_Dai/publication/328271018_Blockchain_challenges_and_opportunities_a_survey/links/5bd2706f92851c6b278f31eb/Blockchain-challenges-and-opportunities-a-survey.pdf. [Accessed Oct. 23, 2018].

[9] Chen, Lin, et al. "On security analysis of proof-of-elapsed-time (poet)." *International Symposium on Stabilization, Safety, and Security of Distributed Systems*. Springer, Cham, 2017, pp282-297.

[10] X. Hao, L. Yu, L. Zhiqiang, L. Zhen and G. Dawu, "Dynamic Practical Byzantine Fault Tolerance," *2018 IEEE Conference on Communications and Network Security (CNS)*, Beijing, 2018, pp. 1-8.

[11] Lesieutre, John, and Z. Wang. "On a generalization of the Collatz conjecture." *MIT*, July, 2007. Available: https://pdfs.semanticscholar.org/9da2/c36caa6a8aca9ed0c3a03b3fe0efd32eb2a8.pdf. [Accessed Oct. 11, 2018].

[12] D. Mingxiao, M. Xiaofeng, Z. Zhe. W. Xiangwei and C. Qijun. "A review on consensus algorithm of blockchain." 2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC), Banff, AB, 2017, pp. 2567-2562. 2017.

[13] Y. Hao, Y. Li, X. Dong, L. Fang and P. Chen, "Performance Analysis of Consensus Algorithm in Private Blockchain," *2018 IEEE Intelligent Vehicles Symposium (IV)*, Changshu, 2018, pp. 280-285.

[14] S. Pongnumkul, C. Siripanpornchana and S. Thajchayapong, "Performance Analysis of Private Blockchain Platforms in Varying Workloads," *2017 26th International Conference on Computer Communication and Networks (ICCCN)*, Vancouver, BC, 2017, pp. 1-6.

[15] Dwork, Cynthia, and Moni Naor. "Pricing via processing or combatting junk mail." *Annual International Cryptology Conference*. Springer, Berlin, Heidelberg, 1992, pp. 139-147.

[16] F. Bocart, "Inflation Propensity of Collatz Orbits: A New Proof-of-Work for Blockchain Applications," *Journal of Risk and Financial Management*, vol. 11, no. 4, p. 83, Nov. 2018.